

**APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY**  
**THIRD SEMESTER B.TECH DEGREE EXAMINATION,**  
**DECEMBER 2020**

**Course Code: BMT203**

**Course Name: DIGITAL ELECTRONICS**

Max. Marks: 100

Duration:  
n: 3  
Hours

**PART A**

*Answer all questions. Each question carries 3 marks*

Marks

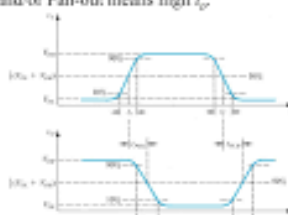
- |   |                              |                                                   |     |
|---|------------------------------|---------------------------------------------------|-----|
| 1 | i)                           | $(3.625)_{10} = (11.101)_2$ (1 Mark)              | (3) |
|   | ii)                          | $(457)_8 = (100101111)_2$ (1 Mark)                |     |
|   | iii)                         | $(102.AB)_{16} = (100000010.10101011)_2$ (1 Mark) |     |
| 2 | Propagation delay (1.5 Mark) |                                                   | (3) |
- Definition: In computer networks, propagation delay is the amount of time it takes for the head of the signal to travel from the sender to the receiver. It can be computed as the ratio between the link length and the propagation speed over the specific medium.
  - Equation and Propagation delay in an inverter 1.5 marks Graph/Wave form

**Propagation Delay**

- **Propagation Delay ( $t_p$ ):** the time needed for the gate to respond to a change at its input. More Fan-In and/or Fan-out means high  $t_p$ .

$$t_p = \frac{t_{pHL} + t_{pLH}}{2}$$

$$f_{max} = \frac{1}{T_{min}} = \frac{1}{2t_p}$$



- **Rise time ( $t_r$ ) and Fall time ( $t_f$ ):** the time difference between 10% of the signal voltage to 90% of signal.

- Can you relate the system maximum speed to the propagation delay?
- Why we need to know the rise time ( $t_r$ )?
- If W, why we have these delays?

- Power dissipation (1.5 Mark)
- Definition and Equation,  $P_D = V_{CC} \times I_{CC}(avg)/n$

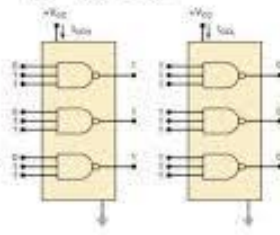
### Digital IC Terminology – Power Requirements

- The amount of power an IC requires is determined by the current,  $I_{CC}$  (or  $I_{DD}$ ) it draws from the supply.
  - Actual power is the product  $I_{CC} \times V_{CC}$  ( $I_{DD} \times V_{DD}$ ).

$$I_{CC(avg)} = \frac{I_{CCH} + I_{CCL}}{2}$$

can be rewritten to calculate average power dissipated:

$$P_D(avg) = I_{CC(avg)} \times V_{CC}$$



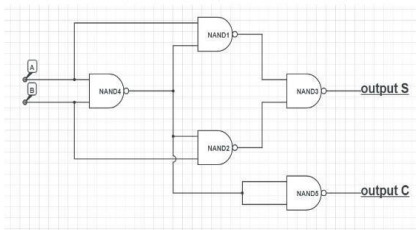
- 3 (i)  $A \cdot B = B \cdot A$  (ii)  $A + B = B + A$  (3)

Law(1 Mark) Proof (2 Marks)

- 4 K map (1.5 Mark) (3)

$F = A\bar{B}$  (1.5 Mark)

- 5 (3)



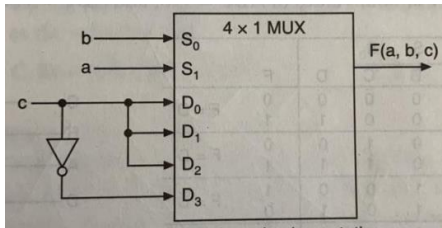
Half adder logic diagram (1.5 Mark)

Sum carry expression and truth table (1.5 Mark)

$$\text{Sum} = A \oplus B$$

$$\text{Carry} = AB$$

6



(3)

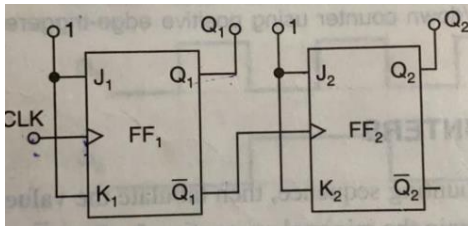
Truth table (1.5 Mark)

Multiplexer implementation (1.5 Mark)

7 SISO, SIPO, PISO, PIPO, Bi directional, Rotate right shift, Rotate left shift register. (1.5 Mark)

Logic representation (1.5 Mark)

8



(3)

(3)

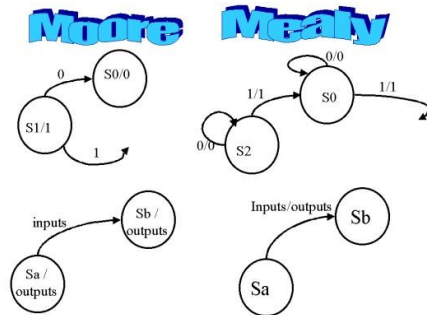
Logic diagram (3 Mark)

9 Steps

(3)

1. Word statement
2. State diagram
3. State table
4. Reduced standard form state table
5. State assignment and transition table
6. Choose type of flip flop and form excitation table
7. K map and minimal expression
8. Realization

10



(3)

Any example with the concept.

## PART B

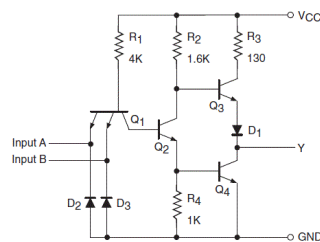
*Answer any one full question from each module. Each question carries 14 marks*

### Module 1

11 Totem pole configuration (7 Marks)

(14)

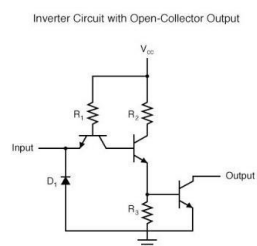
- Logic diagram (3)



- Working, totem pole output, advantage, I/O characteristics, (4)

Open collector gate output(7 Marks)

- Logic diagram (3)



- Working, with external resistor. (4)

12 8421 BCD code (7 Marks)

(14)

- Characteristics

- Invalid states
- Advantages

Unweight self-complementing code. – XS-3 code(7 Marks)

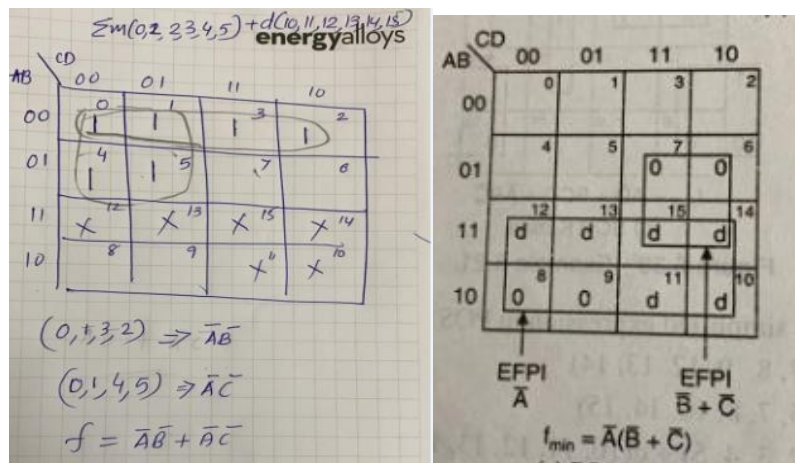
- Characteristics
- Invalid states

## Module 2

13 SOP form- (5 Marks)

(14)

POS form - (5 Marks)



SOP form AOI logic (2 Marks)

POS form AOI logic (2 Marks)

14 Tabular column - (3Marks)

(14)

Prime implicant chart-(3Marks)

Minimal SOP expression- (4 Marks)

K map verification(4 Marks)

| Column 1 |         | Column 2      | Column 3               |
|----------|---------|---------------|------------------------|
| Index    | Minterm | Pairs         | Quads                  |
| Index 0  | 0 ✓     | 0, 1 (1) ✓    | 0, 1, 8, 9 (1, 8) R    |
|          | 1 ✓     | 0, 2 (2) W    | 1, 9, 17, 25 (8, 16) Q |
| Index 1  | 2 ✓     | 0, 8 (8) ✓    | 8, 9, 24, 25 (1, 16) P |
|          | 8 ✓     | 1, 9 (8) ✓    |                        |
|          | 9 ✓     | 1, 17 (16) ✓  |                        |
| Index 2  | 17 ✓    | 8, 9 (1) ✓    |                        |
|          | 24 ✓    | 8, 24 (16) ✓  |                        |
| Index 3  | 21 ✓    | 9, 25 (16) ✓  |                        |
|          | 25 ✓    | 17, 21 (4) V  |                        |
| Index 4  | 15 ✓    | 17, 25 (8) ✓  |                        |
|          | 27 ✓    | 24, 25 (1) ✓  |                        |
| Index 5  | 31 ✓    | 25, 27 (2) U  |                        |
|          |         | 15, 31 (16) T |                        |
|          |         | 27, 31 (4) S  |                        |

$$\begin{aligned}
 P+T+V+W+Q+S &= B\bar{C}\bar{D} + BCDE + A\bar{B}\bar{D}E + \bar{A}B\bar{C}\bar{E} + \bar{C}\bar{D}E + ABDE \\
 P+T+V+W+Q+U &= B\bar{C}\bar{D} + BCDE + A\bar{B}\bar{D}E + \bar{A}B\bar{C}\bar{E} + \bar{C}\bar{D}E + AB\bar{C}\bar{E} \\
 P+T+V+W+R+S &= B\bar{C}\bar{D} + BCDE + A\bar{B}\bar{D}E + \bar{A}B\bar{C}\bar{E} + \bar{A}\bar{C}\bar{D} + ABDE \\
 P+T+V+W+R+U &= B\bar{C}\bar{D} + BCDE + A\bar{B}\bar{D}E + \bar{A}B\bar{C}\bar{E} + \bar{A}\bar{C}\bar{D} + AB\bar{C}\bar{E}
 \end{aligned}$$

### Module 3

- 15 Explain Carry look ahead adder with neat sketches

(14)

(14)

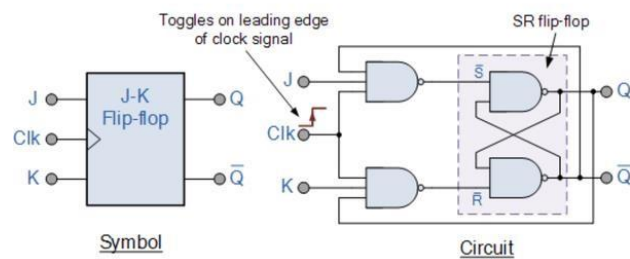
**Derivation 5 marks**

**Explanations 2 marks**

**Circuit diagram 7 marks**

- 16 1. How does a JKFF differ from an SRFF in its basic operation? What is its advantage over an SRFF?

(7)



**Diagrams –2 mark**

| SR Flip Flop Truth Table |   |         |         |
|--------------------------|---|---------|---------|
| R                        | S | Q       | Q'      |
| 0                        | 0 | Q       | Q       |
| 0                        | 1 | 0       | 1       |
| 0                        | 1 | 1       | 1       |
| 1                        | 0 | 0       | 0       |
| 1                        | 0 | 1       | 0       |
| 1                        | 1 | Invalid | Invalid |

| Table 5.4.1 |   |                   |                  |                          |
|-------------|---|-------------------|------------------|--------------------------|
| J           | K | Q Before CK Pulse | Q After CK Pulse | Comments                 |
| 0           | 0 | 0                 | 0                | No Change                |
| 0           | 0 | 1                 | 1                |                          |
| 1           | 0 | 0                 | 1                | $Q = J$<br>$\bar{Q} = K$ |
| 1           | 0 | 1                 | 1                |                          |
| 0           | 1 | 0                 | 0                |                          |
| 0           | 1 | 1                 | 0                |                          |
| 1           | 1 | 0                 | 1                | Outputs Toggle           |
| 1           | 1 | 1                 | 0                |                          |

**Truth tables with explanation 2.5 marks each**

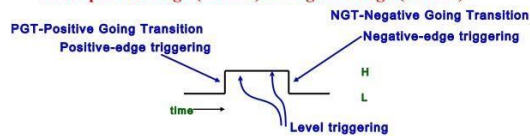
2. Describe the difference between the pulse triggered and edge triggered flipflops.

Difference 3.5 marks

Diagrams 3.5 marks

## TRIGGERING OF FLIP-FLOPS

- *Level-triggering* is the transfer of data from input to output of a flip-flop anytime the clock pulse is proper voltage level.
- *Edge-triggering* is the transfer of data from input to output of a flip-flop on the rising edge (L-to-H) or falling edge (H-to-L) of the clock pulse. Edge triggering may be either positive-edge (L-to-H) or negative-edge (H-to-L).



### Module 4

(14)

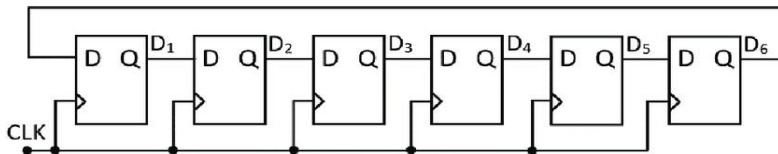
#### 17 Ring counter (7 Marks)

(14)

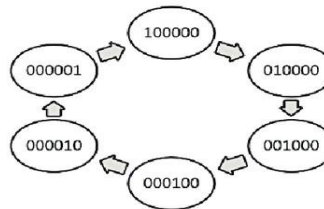
- State diagram
- Truth table
- Logic diagram

#### Twisted ring - Johnson counter (7 Marks)

- State diagram
- Truth table
- Logic diagram

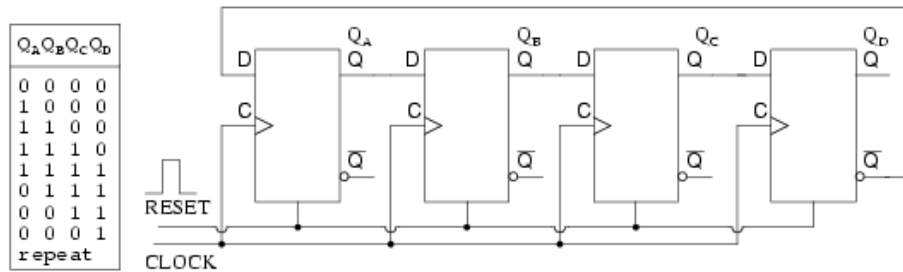


| CLK | D <sub>1</sub> | D <sub>2</sub> | D <sub>3</sub> | D <sub>4</sub> | D <sub>5</sub> | D <sub>6</sub> |
|-----|----------------|----------------|----------------|----------------|----------------|----------------|
| 1   | 1              | 0              | 0              | 0              | 0              | 0              |
| 2   | 0              | 1              | 0              | 0              | 0              | 0              |
| 3   | 0              | 0              | 1              | 0              | 0              | 0              |
| 4   | 0              | 0              | 0              | 1              | 0              | 0              |
| 5   | 0              | 0              | 0              | 0              | 1              | 0              |
| 6   | 0              | 0              | 0              | 0              | 0              | 1              |



Ring counter





Johnson counter (note the  $\overline{Q}_D$  to  $D_A$  feedback connection)

18 Mod 10 asynchronous counter using T flip flop (14)

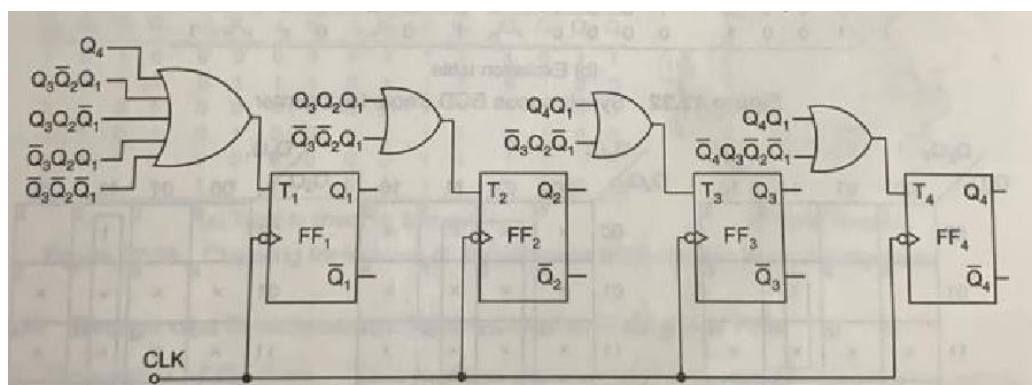
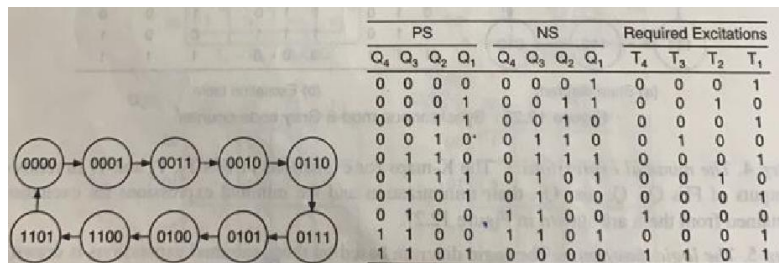
- Count table(4 Marks)
- K map(3 Marks)
- Logic diagram(4 Marks)

Effects of propagation delay in ripple counter (3Marks)

**Module 5** (14)

19 Synchronous Modulo-10 gray code counter. (14)

- State diagram(2 Marks)
- Excitation table(3 Marks)
- K map(6 Marks)
- Logic diagram(3 Marks)



20 Mealy circuit model(5 Marks)

(14)

- Logic diagram
- State table
- State diagram
- Mealy circuit model

Moore circuit model (5 Marks)

- Logic diagram
- State table
- State diagram
- Moore circuit model

Comparison (4 Marks)

| Mealy Machine                                                                                                                         | Moore Machine                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Output depends both upon the present state and the present input                                                                      | Output depends only upon the present state.                                                                                                   |
| Generally, it has fewer states than Moore Machine.                                                                                    | Generally, it has more states than Mealy Machine.                                                                                             |
| The value of the output function is a function of the transitions and the changes, when the input logic on the present state is done. | The value of the output function is a function of the current state and the changes at the clock edges, whenever state changes occur.         |
| Mealy machines react faster to inputs. They generally react in the same clock cycle.                                                  | In Moore machines, more logic is required to decode the outputs resulting in more circuit delays. They generally react one clock cycle later. |